

Maryam Amjad

AI/ML Engineer | Deep Learning | Computer Vision | NLP
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PROFESSIONAL SUMMARY

AI/ML Engineer building production-grade ML systems – from real-time financial prediction pipelines to RAG/knowledge-graph query engines and multi-modal healthcare AI. Strong foundation in **Deep Learning**, **Computer Vision**, and **NLP**, with hands-on experience shipping, evaluating, and deploying models end-to-end on AWS.

CORE SKILLS

Programming: Python, C++, C, SQL

ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn

LLM & GenAI: LangChain, Groq, Hugging Face, Transformers, RAGAS, DeepEval, Mem0, Whisper

Data/NLP/CV: Pandas, NumPy, Matplotlib, OpenCV, NLTK, SpaCy, Tesseract

Databases: PostgreSQL, MySQL, Neo4j, Qdrant, SQLAlchemy

MLOps/DevOps: Docker, Kubernetes, Jenkins, Airflow, MLFlow, CI/CD, DVC, Git

Cloud/Other: AWS, Flask, Streamlit, Selenium, Jupyter Notebook

PROFESSIONAL EXPERIENCE

Junior Machine Learning Engineer

Tensor Labs

Sep 2025 – Present

- Built live crypto prediction pipelines:** Developed and maintained real-time breakout prediction pipelines for Ethereum, Dogecoin, and Solana, processing live market data down to 5-minute candle intervals.
- Tuned rule-based signal logic:** Diagnosed and corrected overly strict/loose signal thresholds and cooldown periods to improve prediction frequency without sacrificing accuracy.
- Resolved data pipeline bugs:** Fixed timestamp and interval synchronization issues in SQLAlchemy-based data pipelines, improving data freshness and eliminating downstream dashboard errors.
- Built text-to-SQL query engine (DiscoverIQ):** Developed a unified natural-language-to-SQL pipeline combining Neo4j knowledge graph retrieval, Qdrant vector search, and LLM-based SQL generation/execution over PostgreSQL.
- Contributed to MedGemma clinical assistant:** Built core modules of a multi-modal medical assistant integrating image diagnostics (MedGemma-4b-it), prescription OCR, and drug interaction checking (TxGemma-9b-chat, FDA API).
- Delivered internal workshop:** Presented a workshop at Tensor Labs on Mem0 (memory-augmented LLMs) and Harness Engineering; authored an article on Harness Engineering and Open Knowledge Format (Medium, forthcoming).

Artificial Intelligence Intern

AIM Lab, Islamabad

Jun 2023 – Aug 2023

- Developed AI-powered web app:** Built a tool that auto-generates slide decks from prompts using NLP and CV.
- Applied transfer learning:** Used pre-trained models for content understanding and image matching.

Lab Demonstrator – Machine Learning

FAST NUCES Islamabad

Sep 2024 – Dec 2024

- Conducted labs:** Led weekly labs on feature engineering, hyperparameter tuning, and live ML demos.

PUBLICATIONS & TALKS

“Harness Engineering and Open Knowledge Format for AI Agents”: Medium article – forthcoming

Workshop: Mem0 & Harness Engineering for AI Agents: Tensor Labs, internal

Workshop: LLM Evaluation – Frameworks, Metrics & Tools: Tensor Labs, internal

KEY PROJECTS

Eth Breakout:

Real-time breakout prediction system for Ethereum, Dogecoin, and Solana using rule-based and ML signal logic across 15-min to 5-min candle timeframes; also built a liquidation/liquidity heatmap visualizing leveraged position clusters from open interest, funding rate, and live price data. *Technologies: Python, SQLAlchemy, Pandas, Scikit-learn, PostgreSQL, WebSockets, Streamlit*

DiscoverIQ – Text-to-SQL & Knowledge Graph Query Engine:

Unified natural-language query pipeline that converts chat messages into executable SQL, combining knowledge graph context retrieval with schema-aware SQL generation and execution. *Technologies: Python, Neo4j, Qdrant, PostgreSQL, LangChain, Groq LLM*

MedGemma Multi-Modal Clinical Assistant:

Full-stack clinical assistant combining medical image diagnostics, prescription/medicine OCR, drug interaction checking, and personal health tracking in one platform. *Technologies: MedGemma-4b-it, Whisper-large-v3, TxGemma-9b-chat, FDA API, Python, Flask*

LLM Evaluation Toolkit:

Hands-on toolkit covering 12 LLM evaluation categories (text generation, summarization, RAG, text-to-SQL, safety/robustness, long context, multi-modal, and more) with runnable notebooks and an interactive Streamlit dashboard; open-sourced on GitHub and used as the basis for an internal workshop. *Technologies: Python, Groq, RAGAS, DeepEval, HuggingFace Evaluate, PromptBench, Streamlit*

Straight Egyptian Arabian Horse Knowledge Graph:

Interactive, fully Dockerized web app for exploring horse pedigrees, show results, stables, and bloodlines through a Neo4j graph database, with 5-generation interactive pedigree visualization, top-10 rankings, and search/filtering. *Technologies: Python, Pandas, Neo4j, Streamlit, PyVis, Docker*

TherapEase – Autism Therapy Assistant (Final Year Project):

3D digital twin with real-time emotion detection for autism therapy support. *Technologies: React, Three.js, OpenCV, MediaPipe, Flask, TensorFlow, DeepFace*

EDUCATION

B.S. in Artificial Intelligence

FAST NUCES, Islamabad

Relevant Coursework: Deep Learning, Generative AI, CV, NLP

Jul 2021 - Jun 2025

CERTIFICATIONS & AWARDS

AWS Certified AI Practitioner (AWS, Passed March 2026) • Generative AI with LLMs (DeepLearning.AI, 2025) • AI Agents with RAG and LangChain (IBM, 2025) • Convolutional Neural Networks (DeepLearning.AI, 2024) • AWS Cloud Essentials (AWS, 2025)